# THE WORLD THROUGH SHATTERED GLASS AND ROSE TINTS

## ALGORITHMIC BIAS AND DISCRIMINATION - HOW HISTORICAL INEQUALITIES CAN SHAPE TODAY'S CREDIT DESCISIONS

DATA DRIVEN DESCISIONS POWERED BY:
PYTHON | PANDAS | NUMPY | MATPLOTLIB

## THE ANALOGY

Imagine viewing the world through a pair of glasses, shattered but painted in a rose coloured tint.

- The cracks are the legacies of apartheid; a broken distribution of income and wealth and fractured access to education and employment.
- The rose tint is the monetisable gain from a bias within the data.
- The result is a distorted, rosy picture that flatters some and marginalises others.

## A BEAUTIFULLY FLAWED MODEL

## OPPORTUNITIES IN PROXYING

## LEARNED BIASES

## REINFORCES THE BIAS

## DISCRIMINATORY TARGETING

## HOW PROXY DISCRIMINATION ARISES

- CreditScoreCheck Ltd scores applicants using income,housing and employment stability
- variables are racially skewed due to apartheid's structural legacy
- Black applicants score higher risk, not because they are, but because history said so
- Biased profiles are sold to lenders as objective data products
- Denied applicants cannot build credit history and the cycle continues

## SUMMARY

This project examined demographic bias in credit approval profiling at CreditScoreCheck Ltd, a financial data brokerage in the SADC region. Using Python, Pandas, NumPy and Matplotlib, the team merged a real-world Kaggle credit approval dataset comprising application records and credit repayment histories. Repayment statuses were used to derive a binary risk outcome, categorising high and low risk profiles. A synthetic race variable was generated using Stats SA demographic proportions, assigned by income brackets, reflecting South Africa's historically unequal income distribution.

A proxy-based risk scoring framework was established using income, employment stability, housing and education. This revealed a 17.8% approval gap between Black and White applicants at approval rates of 66.3% and 84.1% respectively and a 0.788 Disparate Impact Ratio, which fell below the 0.800 legal fairness threshold. Race was never an explicit input, however, the disparity surfaced through proxy variables, demonstrating how historical inequality is unintentionally carved into today's decisions.

Vallor & Rewark (2018) | Bostrom & Yudkowsky (2018) | Jordon et al. (2022) | Infosys (2025) | Stats SA (2022)

Confidential — Internal Ethics Review